Datasets:

JMulTi http://www.jmulti.de/data_atse.html Applied Time Series Econometrics – Helmut Lutkepohl

German GNP series: the break date is known to be the third quarter of 1990
U.S. investment and German interest rate series. In no case can the stationarity null hypothesis be rejected at the 5% level for the U.S. investment series, whereas it is rejected for the German interest rate. This result corresponds quite nicely to rejecting the unit root for the U.S. investment series and not rejecting it for the interest rate with the ADF and Schmidt–Phillips tests. Notice, however, that using lq = 12 for the investment series results in a test value that is significant at the 10% level. Still, taking all the evidence together, we find that the series is better viewed as I(0) than as I(1).

seasonally adjusted, quarterly German consumption data for the period 1960Q1 − 1982Q4. The series is the consumption series given in Table E.1 of Lutkepohl (1991). They reveal that constructing a model for the logs is likely to be advantageous because the changes in the log series display a more stable variance than the changes
in the original series. At the 5% level the unit root cannot be rejected. The corresponding KPSS test, again with a constant and a time trend, confirms the result. It clearly rejects stationarity at the 5% level. **This outcome results with a lag truncation parameter of three, but a rejection of stationarity is also obtained with other lag orders. Overall, the test results support one unit root in the log consumption series; thus, specifying a stationary model for the first differences seems appropriate.**

The second example series consists of the logarithms of a seasonally unadjusted quarterly Polish productivity series for the period 1970Q1–1998Q4. Although the series is not seasonally adjusted, it does not have a very clear seasonal pattern. the presence of a structural shift in 1990Q1 for Polish productivity. case the unit root is clearly rejected even at the 1% level. Hence, we continue the analysis with the first differences of the series and include an impulse dummy as a deterministic term in addition to a constant.

https://www.time-series.net/data_sets - Applied Econometric Time Series – Walter Enders

Wei: W1 series: stationary

Detected Outliers

Iteration	Time	Type
1	36	AO
2	9	IO
3	7	AO
4	4	IO

W2 series stationary in mean but may not in variance

W3: series stationary in mean but may not in variance

Blow-fly data: typo at t=20

W4: non stat in mean

Detected Outliers		
Iteration	Time	Type
1	81	TC
2	82	TC

W5: non stat in mean with increasing trend

W6: non stat in mean and variance

W7: log series is stat

W10 - seasonal quarterly – seasonal unit root

- -

Detected Outliers			
Iteration	Time	Type	Magnitude ($\hat{\omega}$)
1	12	AO	16.26
2	27	IO	−2.31

Monthly O3 readings: January 1960 break

EXAMPLE 10.3 The Arab oil embargo in November 1973 greatly affected the supply of petroleum products to the United States. The prices of petroleum products rocketed, and the cries for energy conservation were loud and clear everywhere in the United States. It is generally believed that the rate of growth of energy consumption following this event was lower for the post-embargo years. To test the validity of this belief, Montgomery and Weatherby (1980) applied an intervention model to the natural logarithm of monthly electricity consumption from January 1951 to April 1977 using a total of 316 observations.

Although the embargo started in November 1973, Montgomery and Weatherby (1980) assumed that the effect of this embargo was not felt until December 1973. Thus, they used 275 months of data from January 1951 to November 1973 to model the noise series and obtained the following ARIMA$(0, 1, 2) \times (0, 1, 1)_{12}$ model:

Shumway & Stoffer: Time Series Analysis and Its Applications With R Examples – 3rd edition

Twenty-four month forecasts for the Recruitment series. The actual data shown are from about January 1980 to September 1987 – stationary

In this example, we consider the analysis of quarterly U.S. GNP from 1947(1) to 2002(3), n = 223 observations. The data are real U.S. gross national product in billions of chained 1996 dollars and have been seasonally adjusted. The data were obtained from the Federal Reserve Bank of St. Louis (http://research.stlouisfed.org/). the trend has been removed we are able to notice that the variability in the second half of the data is larger than in the first half of the data. Also, it appears as though a trend is still present after differencing. ARIMA(1; 1; 0) for log GNP.

the Monthly Federal Reserve Board Production Index and Unemployment (1948-1978, n = 372 months). ARIMA(0; 1; 1) (0; 1; 1)12

Periodogram of SOI and Recruitment, n = 453 (n0 = 480), where the frequency axis is labeled in multiples of = 1=12. one cycle per year (12 months), one cycle every four years (48 months).

Brockwell and Davis Time Series: Theory and Methods 1991 Brockwell & Davis Data Files http://users.stat.umn.edu/~kb/classes/5932/BDFiles.html

Population of the U.S.A. at ten-year intervals, 1 790- 1980 - the second to a roughly exponentially increasing graph,

the third to a graph which fluctuates erratically about a nearly constant or slowly rising level, Strikes in the U.S.A., 1 95 1 - 1 980 (Bureau of Labor Statistics, U.S. Labor Department).

the fourth to an erratic series of minus ones and ones. All-star baseball games, 1933 - 1 980.

The fifth graph appears to have a strong cyclic component with period about 11 years The Wolfer sunspot numbers, 1770- 1869.

and the last has a pronounced seasonal component with period 12. Monthly accidental deaths in the U.S.A., 1 973 - 1 978 (National Safety Council). The Small Trend

International airline passengers; monthly totals in tlrousands of passengers { U, t = I , . . . , 144} from January 1949 to December 1960 (Box and Jenkins (1970)). Regular and seasonal stochastic trend

Peter J. Brockwell Richard A. Davis – Introduction to Time Series and Forecasting Second Edition

Monthly accidental deaths regular and seasonal unit root

Bai-Perron: Bai, J. and Perron, P. (2003), Computation and analysis of multiple structural change models. J. Appl. Econ., 18: 1-22. https://doi.org/10.1002/jae.659

US ex-post real interest rate 1961:1–1986:3 The break dates are estimated at 1966:4, 1972:3 and 1980:3.

POST-war UK inflation rate 1947–1987 The first break date is the same as in the one-break model, namely 1967, which is linked to the end of the Bretton Woods system. The second break is located in 1975.

Eric Zivot and Donald W. K. Andrews, Further Evidence on the Great Crash, the Oil-Price Shock, and the Unit-Root Hypothesis, Journal of Business & Economic Statistics , Jul., 1992, Vol. 10, No. 3 (Jul., 1992),pp. 251-270

jstor.org/stable/pdf/1391541.pdf?refreqid=fastly-default%3A12d20c0e8bcfe309f5aa55679e859d51&ab_segments=&initiator=&acceptTC=1

Table 1. Minimum t Statistics

Series^i (i = A, B, C)	Rank 1 *t stat*	Year	2 *t stat*	Year	3 *t stat*	Year
Real GNP[A]	−5.58***	1929	−4.34	1928	−3.89	1927
Nominal GNP[A]	−5.82***	1929	−4.36	1927	−4.23	1928
Real per capita GNP[A]	−4.61***	1929	−4.29	1928	−4.09	1927
Industrial production[A]	−5.95***	1929	−5.40	1928	−5.09	1927
Employment[A]	−4.95***	1929	−4.71	1928	−4.38	1927
GNP deflator[A]	−4.12**	1929	−3.90	1928	−3.82	1930
Consumer prices[A]	−2.76	1873	−2.69	1872	−2.64	1864
Nominal wages[A]	−5.30***	1929	−5.10	1930	−4.61	1928
Money stock[A]	−4.34***	1929	−4.34	1928	−4.32	1930
Velocity[A]	−3.39	1949	−3.35	1947	−3.21	1946
Interest rate[A]	−.98	1932	−.96	1965	−.91	1967
Quarterly real GNP[B]	−4.08**	1972:II	−4.07	1972:III	−4.07	1972:I
Common-stock prices[C]	−5.61***	1936	−5.60	1937	−5.52	1939
Real wages[C]	−4.74***	1940	−4.67	1941	−4.59	1931

NOTE: The minimum t statistics were determined as follows. For each series, Equation (1′), (2′), or (3′) was estimated with the breakpoint, T_B, ranging from $t = 2$ to $t = T$-1. For each regression, k was determined as described in the paragraph following Equation (3′), and the t statistic for testing $\alpha^i = 1$ was computed. The minimum t statistic reported is the minimum over all T-2 regressions. The symbols *, **, and *** indicate that the unit-root hypothesis is rejected at the 10%, 5%, and 1% levels, respectively, using Perron's critical values.

Table 6. Tests for a Unit Root

Series	T	$\hat{T}_B$	k		$\hat{\mu}^A$	$\hat{\theta}^A$	$\hat{\beta}^A$	$\hat{\alpha}^A$	$S(\hat{e})$
Model A: Regression: $y_t = \hat{\mu}^A + \hat{\theta}^A DU(\hat{\lambda})_t + \hat{\beta}^A t + \hat{\alpha}^A y_{t-1} + \Sigma_1^k \hat{c}_j^A \Delta y_{t-j} + \hat{e}_t$									
Real GNP	62	1929	8		3.514	−.195	.027	.267	.05
					(5.62)	(−4.92)	(5.71)	(−5.58)***	
Nominal GNP	62	1929	8		5.040	−.311	.032	.532	.07
					(5.85)	(−5.12)	(5.97)	(−5.82)***	
Real per capita GNP	62	1929	7		3.584	−.117	.012	.494	.056
					(4.62)	(−3.41)	(4.69)	(−4.61)*	
Industrial production	111	1929	8		.122	−.317	.034	.290	.088
					(4.46)	(−5.12)	(5.91)	(−5.95)***	
Employment	81	1929	7		3.564	−.051	.006	.651	.029
					(4.97)	(−3.14)	(4.79)	(−4.95)**	
GNP Deflator	82	1929	5		.641	−.091	.007	.786	.044
					(4.17)	(−3.23)	(4.14)	(−4.12)	
Consumer prices	111	1873	2		.217	−.055	.001	.941	.043
					(2.79)	(−2.51)	(3.27)	(−2.76)	
Nominal wages	71	1929	7		2.126	−.161	.017	.660	.054
					(5.35)	(−4.16)	(5.32)	(−5.30)**	
Money stock	82	1929	6		.288	−.064	.011	.823	.044
					(4.76)	(−2.54)	(4.25)	(−4.34)	
Velocity	102	1949	0		.224	.095	−.002	.840	.064
					(2.99)	(3.09)	(−2.95)	(−3.39)	
Interest rate	71	1932	2		.065	−.444	.013	.945	.272
					(.31)	(−2.55)	(3.09)	(−.98)	
Model B: Regression: $y_t = \hat{\mu}^B + \hat{\beta}^B t + \hat{\gamma}^B DT^*(\hat{\lambda})_t + \hat{\alpha}^B y_{t-1} + \Sigma_1^k \hat{c}_j^B \Delta y_{t-j} + \hat{e}_t$									
Quarterly real GNP	159	72:II	10		1.044	.001	−.0004	.851	.010
					(4.12)	(3.93)	(−2.86)	(−4.08)	
Model C: Regression: $y_t = \hat{\mu}^C + \hat{\theta}^C DU(\hat{\lambda})_t + \hat{\beta}^C t + \hat{\gamma}^C DT^*(\hat{\lambda})_t + \hat{\alpha}^C y_{t-1} + \Sigma_1^k \hat{c}_j^C \Delta y_{t-j} + \hat{e}_t$									
Common-stock prices	100	1936	1	.471	−.226	.007	.021	.642	.139
				(5.12)	(−3.25)	(4.83)	(4.80)	(−5.61)***	
Real wages	71	1940	8	2.678	.085	.012	.008	.115	.030
				(4.81)	(4.33)	(4.49)	(3.68)	(−4.74)	

NOTE: t statistics are in parentheses. The t statistic for $\hat{\alpha}^i$ is for testing $\alpha^i = 1$. k is determined as described in the paragraph following Equation (3′). The symbols *, **, and *** denote significance of the test of $\alpha^i = 1$ at the 10%, 5%, and 1% levels, respectively, using the critical values from Table 2A, 3A, or 4A.

Chow test:
https://web.archive.org/web/20191228155733/http://pdfs.semanticscholar.org/0f70/219160c8ad2f9db02e226d3f7d7320e729b8.pdf

RegressionsGregory C. Chow, Tests of Equality Between Sets of Coefficients in Two Linear, Econometrica, Vol. 28, No. 3. (Jul., 1960), pp. 591-605.

X_t, ownership of automobiles measured in "new-car equivalents" per capita at the end of year t. The unit is per cent of a new-car equivalent per capita.

X_t^1, purchase of new cars during year t, with the same unit of measurement as above.

P_t, relative price index of automobile stock, with 1937 as 100.

I_{dt}, real disposable income per capita in 1937 dollars.

I_{et}, real "expected" income per capita in 1937 dollars used by Milton Friedman in his *A Theory of the Consumption Function* (Princeton: Princeton University Press, 1957).

A statistical demand function for automobile ownership computed from observations of 33 years from 1921 to 1953 is

$$\text{(X1e)} \qquad \hat{X}_t = -.7247 - \underset{(.004201)}{.048802}\, P_t + \underset{(.001747)}{.025487}\, I_{et}, \qquad R^2 = .895, \quad s = .618.$$

A statistical demand function for new purchase computed from observations of 28 years, from 1921 to 1953 but excluding 1942 to 1946, is

$$\text{(4s)} \qquad \hat{X}_t^1 = .07791 - \underset{(.002648)}{.020127}\, P_t + \underset{(.001070)}{.011699}\, I_{dt} - \underset{(.04719)}{.23104}\, X_{t-1}, \quad R^2 = .858, \quad s = .308.$$

	1954	1955	1956	1957
X_t estimated from (X1e)	12.665	12.993	13.328	13.025
X_t observed minus estimated	—.613	.079	.102	.437

The residuals of the observed values from the estimated values are very small, as compared with the standard error of .618. They do not indicate any shifts in the pattern of demand for automobile ownership during the four years 1954 to 1957.

https://arxiv.org/pdf/2601.02957v3

Datasets. We curate seven benchmark datasets from the Turing Change Point Dataset and other public sources, each with a single, historically documented structural break whose cause is unambiguous (Table 1). These datasets span diverse domains and vary in length (21–468 observations), frequency (annual to monthly), and change characteristics (mean shifts, trend changes, variance changes).

Dataset	**Domain**	**N**	**Event**
Nile	Hydrology	100	Aswan Dam (1898)
Seatbelts	Policy	108	UK Law (1983)
LGA	Aviation	468	9/11 (2001)
Ireland Debt	Economics	21	Banking Crisis (2009)
Ozone	Environment	54	Peak Depletion (1993)
Robocalls	Telecom	53	FCC Ruling (2018)
Japan Nuclear	Energy	40	Fukushima (2011)

Table 1: Benchmark datasets with ground truth structural breaks.

The datasets represent different types of realworld changepoints: Nile (annual river flow, 1871–1970) shows a mean shift when the Aswan Low Dam construction began in 1898 (Cobb, 1978); Seatbelts (monthly UK road casualties, 1976– 1984) exhibits a sudden drop following the 1983

compulsory seatbelt law (Harvey and Durbin, 1986); LGA (monthly LaGuardia Airport passengers, 1977–2015) captures the immediate and sustained impact of the September 11 attacks on air travel (Ito and Lee, 2005); Ireland Debt (annual debt-to-GDP ratio, 2000–2020) shows the dramatic surge following the 2008 banking crisis and subsequent bailout (Lane, 2011); Ozone (annual Antarctic zone measurements, 1961–2014) marks the reversal point when Montreal Protocol effects began (Solomon et al., 2016); Robocalls (monthly US call volume, 2015–2019) increases sharply after a 2018 federal court ruling loosened FCC restrictions; and Japan Nuclear (annual nuclear share of electricity, 1985–2024) drops precipitously after the 2011 Fukushima disaster led to reactor shutdowns (Hayashi and Hughes, 2013). Most datasets are sourced from the Turing Change Point Dataset (van den Burg and Williams, 2020). https://github.com/alan-turing-institute/TCPD/tree/master/datasets